Hints: Trigonometry – 2. Roots of unity and the complex exponential

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Put $\zeta = e^{2\pi i/p}$ and write $|G|^2 = G\bar{G} = \sum_{j,k} \zeta^{j^2-k^2} = \sum_{j,k} \zeta^{(j-k)(j+k)}$.

Hint 2. Since p is odd, $(j,k) \mapsto (u,v) = (j-k, j+k)$ is a bijection of $(\mathbb{Z}/p)^2$. So $|G|^2 = \sum_u \sum_v \zeta^{uv}$, and by the root-of-unity filter the inner sum is p for $u = 0$ and 0 otherwise.

Hint 3. Each nonzero square is hit by two values of k, so $G = \sum_{a=0}^{p-1}(1 + (\frac{a}{p}))\zeta^a = \sum_a (\frac{a}{p})\zeta^a$. Substituting $a \to -a$ gives $\bar{G} = (\frac{-1}{p})G$, hence $G^2 = (\frac{-1}{p})|G|^2 = (\frac{-1}{p})\, p = (-1)^{(p-1)/2}p$.

Problem 2

Hint 1. Put $x_j = 2\cos\frac{2\pi j}{7}$, $j = 1,2,3$. From $z^7 = 1$, $z \neq 1$, i.e. $z^3+z^2+z+1+z^{-1}+z^{-2}+z^{-3} = 0$, with $x = z+z^{-1}$ one gets that x_1, x_2, x_3 are the roots of $P(x) = x^3+x^2-2x-1$, which is irreducible over $\mathbb{Q}$ (no rational roots). Since $2\cos 2\theta = x^2 - 2$ and $2\cos 3\theta = x^3 - 3x$ for $x = 2\cos\theta$, there is $R \in \mathbb{Q}[x]$ with $R(2\cos\theta) = a + b\cos\theta + c\cos 2\theta + d\cos 3\theta$ for all θ; the right-hand side of the problem is $R(x_1)$.

Hint 2. Suppose $R(x_1)^N = 2$, where $N = 3$ or 5. Then P, being irreducible with the root x_1, divides $R^N - 2$, so $R(x_2)$ and $R(x_3)$ are also real N-th roots of 2. As N is odd, $R(x_1) = R(x_2) = R(x_3) = \sqrt[N]{2}$.

Hint 3. Add the three equalities: $3\sqrt[N]{2} = \sum_{j=1}^{3}(a + b\cos\frac{2\pi j}{7} + c\cos\frac{4\pi j}{7} + d\cos\frac{6\pi j}{7}) = 3a - \frac{b+c+d}{2}$, because by the root-of-unity filter $\sum_{j=1}^{3}\cos\frac{2\pi jm}{7} = \frac{1}{2}\sum_{j=1}^{6}\cos\frac{2\pi jm}{7} = -\frac{1}{2}$ for $7 \nmid m$. This contradicts the irrationality of $\sqrt[N]{2}$. Answer: no in both cases.

Hint 4. Where the answers come from: the numbers $1, \cos\frac{2\pi}{7}, \cos\frac{4\pi}{7}, \cos\frac{6\pi}{7}$ span the field $K = \mathbb{Q}(\cos\frac{2\pi}{7})$ of degree 3. For $\sqrt[5]{2}$ (degree $5 \nmid 3$) the degree alone already forbids $\sqrt[5]{2} \in K$. For $\sqrt[3]{2}$ the degrees agree, and the obstruction is the one used above: together with x_1 the field K contains all its conjugates x_2, x_3 (it is a normal, cyclic extension), whereas $\mathbb{Q}(\sqrt[3]{2})$ does not contain the non-real conjugates of $\sqrt[3]{2}$.

Problem 3

Hint 1. From $\cos(n+1)\theta + \cos(n-1)\theta = 2\cos\theta\cos n\theta$ get $T_{n+1} = 2xT_n - T_{n-1}$, $T_0 = 1$, $T_1 = x$; induction gives integer coefficients and leading coefficient 2^{n-1} (for $n \geqslant 1$).

Hint 2. $D_n(x) = 2T_n(x/2)$ satisfies $D_{n+1} = xD_n - D_{n-1}$, $D_0 = 2$, $D_1 = x$, so for $n \geqslant 1$ it is monic with integer coefficients; and $D_n(2\cos\theta) = 2\cos n\theta$. Hence $s = 2\cos\frac{2\pi}{n}$ is a root of the monic integer polynomial $D_n(x) - 2$: an algebraic integer.

Hint 3. For the degree (for $n > 2$): $\varepsilon = e^{2\pi i/n}$ has degree $\varphi(n)$ (irreducibility of Φ_n), and it satisfies $\varepsilon^2 - s\varepsilon + 1 = 0$ over $\mathbb{Q}(s)$ but is not real. Hence $[\mathbb{Q}(\varepsilon) : \mathbb{Q}(s)] = 2$ and $[\mathbb{Q}(s) : \mathbb{Q}] = \varphi(n)/2$.

Problem 4

Hint 1. $\arctan k = \arg(1 + ki)$, so $e^{2i\arctan k} = \frac{1+ki}{1-ki}$. If the identity held, then $\prod_{k=1}^{15}(\frac{1+ki}{1-ki})^{m_k} = \frac{1+16i}{1-16i}$. Moving the factors with negative exponents to the other side, get an equality $(1+16i)\,X = (1-16i)\,Y$, where X, Y are products of numbers $1 \pm ki$, $1 \leqslant k \leqslant 15$.

Hint 2. $\pi = 1 + 16i$ has norm 257, a prime, so π is a Gaussian prime. It divides the left side. Does it divide the right side?

Hint 3. If $\pi \mid 1 \pm ki$, then $257 = N(\pi)$ divides $N(1 \pm ki) = 1 + k^2 \leqslant 226$, impossible; and $\pi \mid 1 - 16i$ would give $\pi \mid (1+16i) + (1-16i) = 2$, impossible since $257 \nmid 4$. By unique factorisation in $\mathbb{Z}[i]$, π does not divide $(1 - 16i)Y$: a contradiction. Answer: no such integers exist.

Problem 5

Hint 1. $|1 - \varepsilon^k| = 2 \sin \frac{\pi k}{n}$, so you need $\sum_{k=0}^{n-1} \sin \frac{\pi k}{n} = \operatorname{Im} \sum_{k=0}^{n-1} e^{i\pi k/n}$.

Hint 2. The geometric sum equals $\frac{1-e^{i\pi}}{1-e^{i\pi/n}} = \frac{2}{1-e^{i\pi/n}}$, and $1 - e^{i\alpha} = -2i \sin \frac{\alpha}{2}\, e^{i\alpha/2}$; its imaginary part is $\cot \frac{\pi}{2n}$.

Hint 3. Interpretation: $n|1 - \varepsilon^k|$ is the perimeter of the star polygon $\{n/k\}$ obtained by joining every k-th vertex of the regular n-gon. So the sum is $\frac{1}{n}$ times the total perimeter of all star polygons $\{n/k\}$, $1 \leqslant k \leqslant n - 1$ (twice the total length of all sides and diagonals, divided by n).

Problem 6

Hint 1. On $|z| = 1$ we have $\bar{z} = 1/z$, so $1 = p(z)\overline{p(z)} = p(z)\, \bar{p}(1/z)$, where $\bar{p}$ has the conjugated coefficients. Multiply by z^n, $n = \deg p$, to get rid of negative powers.

Hint 2. With $p^*(z) = z^n \bar{p}(1/z)$ (a polynomial) the identity $p(z)p^*(z) = z^n$ holds on the unit circle, hence everywhere. So p divides z^n, i.e. $p = cz^m$, and $|c| = 1$.

Hint 3. A trigonometric polynomial is $f(\theta) = e^{-iN\theta} p(e^{i\theta})$ for a polynomial p; apply the first part to p.

Problem 7

Hint 1. Pair k with $n - k$: for $|w| = 1$, $w \neq 1$, one has $\frac{1}{1-w} + \frac{1}{1-\bar{w}} = 1$. Hence $\sum_{k=1}^{n-1} \frac{1}{1-\varepsilon^k} = \frac{n-1}{2}$.

Hint 2. $\frac{\varepsilon^k}{1-\varepsilon^k} = \frac{1}{1-\varepsilon^k} - 1$, so the first sum is $-\frac{n-1}{2}$.

Hint 3. $\frac{1}{1-e^{i\alpha}} = \frac{1}{2} + \frac{i}{2} \cot \frac{\alpha}{2}$. Taking imaginary parts in the second sum gives $\sum_{k=1}^{n-1} \cot \frac{k\pi}{n} = 0$ (also clear from the symmetry $k \leftrightarrow n - k$).

Problem 8

Hint 1. Start with the product formula: every root of $z^n - 1$ is a primitive d-th root of unity for exactly one $d \mid n$, and the primitive d-th roots are $e^{2\pi i k/d}$ with $(k, d) = 1$. Grouping the factors of $z^n - 1 = \prod_k (z - \varepsilon^k)$ by order gives $\prod_{d|n} \Phi_d = z^n - 1$.

Hint 2. Integrality by strong induction on n: $\Phi_n = (z^n - 1) / \prod_{d|n, d<n} \Phi_d$, and division by a monic integer polynomial keeps integer coefficients.

Hint 3. $\Phi_p = 1 + z + \cdots + z^{p-1}$; $\Phi_{p^2} = (z^{p^2} - 1)/(z^p - 1) = \Phi_p(z^p)$; for odd p, $\Phi_{2p}(z) = \Phi_p(-z)$ (the map $\zeta \mapsto -\zeta$ swaps primitive p-th and $2p$-th roots), while $\Phi_4 = z^2 + 1$.

Problem 9

Hint 1. Split the sum into $k \geqslant 0$ and $k < 0$: it equals $\operatorname{Re}\Big(1 + 2\sum_{k\geqslant 1} (re^{i\theta})^k\Big) = \operatorname{Re} \frac{1+re^{i\theta}}{1-re^{i\theta}}$. Multiply numerator and denominator by $1 - re^{-i\theta}$.

Hint 2. For the series: $\sum_{k\geqslant 1} \frac{r^k \cos k\theta}{k} = \operatorname{Re} \sum_{k\geqslant 1} \frac{(re^{i\theta})^k}{k} = -\operatorname{Re} \log(1 - re^{i\theta}) = -\frac{1}{2} \ln(1 - 2r\cos\theta + r^2)$ (equivalently, integrate $\frac{P_r(\theta)-1}{2r}$ in r).

Hint 3. Let $r \to 1^-$: the series $\sum \frac{\cos k\theta}{k}$ converges for $\theta \in (0, 2\pi)$ (Dirichlet's test), so Abel's theorem applies, and $-\frac{1}{2}\ln(2 - 2\cos\theta) = -\ln(2\sin\frac{\theta}{2})$.

Problem 10

Hint 1. Expand $|z - \varepsilon^k|^2 = |z|^2 + 1 - 2\,\mathrm{Re}(\bar{z}\varepsilon^k)$ and use $\sum_k \varepsilon^k = 0$.

Hint 2. $\prod_k |z - \varepsilon^k| = |z^n - 1| \leqslant |z|^n + 1 = 2$ on $|z| = 1$, with equality iff $z^n = -1$. The maximum is 2.